HUMAnN3 Functional Profiling with Differential Analysis

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Overview

This practical introduces students to HUMAnN3-derived functional microbiome outputs using pathway abundance, pathway coverage, and gene family tables.

Questions addressed in this session

1. What microbial functions are most abundant across the cohort?
2. Do samples cluster by clinical group based on functional profiles?
3. Which broad biological categories dominate the microbiome?
4. Which pathways differ across groups?
5. Which gene families differ across groups?
6. Do the results highlight biology linked to the paper, such as metabolism, respiration, iron/heme handling, and nutrient utilization?

Expected learning points

- HUMAnN outputs include both broad functional summaries and high-resolution gene family features.
- Pathway abundance and pathway coverage answer different questions.
- Gene-family analysis requires filtering because of the very large number of features.
- Exploratory plots show broad structure, while differential abundance analysis identifies specific group-associated functions.

Prerequisites

This module assumes:

- Basic familiarity with R and RStudio
- Completed upstream processing (e.g., BWA, Prokka, Sylph, MetaPhlAn, HUMAnN3)
- Availability of:
  - pathway abundance tables
  - pathway coverage tables
  - gene family tables
  - sample metadata

Required R packages: - tidyverse - pheatmap - vegan - Maaslin3

Input data structure

- Rows = features (pathways or gene families)
- Columns = samples
- First column = Feature IDs
- Metadata must include:
  - sample_name
  - run_accession
  - group

```r
# ============================================
# LIBRARIES
# ============================================

library(tidyverse)
library(pheatmap)
library(vegan)

if (!requireNamespace("maaslin3", quietly = TRUE)) {
  if (!requireNamespace("BiocManager", quietly = TRUE)) {
    install.packages("BiocManager")
  }
  BiocManager::install("maaslin3")
}
library(maaslin3)
```

```r
# ============================================
# HELPER-FUNCTIONS
# ============================================

extract_run_id <- function(x) {
  x <- gsub("_Abundance-RPKs$", "", x)
  x <- gsub("_Abundance$", "", x)
  x <- gsub("_Coverage$", "", x)
  x
}

clean_names <- function(x) {
  x <- gsub("^.*?:\\s*", "", x)   # remove pathway ID prefix
  x <- gsub("\\(.*?\\)", "", x)   # remove bracketed text like (E. coli)
  x <- gsub("_", " ", x)
  trimws(x)
}

annotate_pathways <- function(x) {
  case_when(
    grepl("glycolysis|fermentation|respiration|TCA|energy|formate", x, ignore.case = TRUE) ~ "Energy met
    grepl("amino acid|valine|leucine|isoleucine|ornithine|biosynthesis", x, ignore.case = TRUE) ~ "Amino
    grepl("nucleotide|purine|pyrimidine|UMP|adenine|guanosine|allantoin", x, ignore.case = TRUE) ~ "Nucl
    grepl("peptidoglycan|cell wall|muramoyl|lipopolysaccharide", x, ignore.case = TRUE) ~ "Cell wall",
    grepl("carbohydrate|glycogen|sugar|fucose|galactarate", x, ignore.case = TRUE) ~ "Carbohydrate metal
    grepl("cofactor|vitamin|heme|iron|aerobactin|ubiquinol|menaquinol", x, ignore.case = TRUE) ~ "Cofact
    TRUE ~ "Other"
```

```r
  )
}

remove_zero_variance_features <- function(mat) {
  keep <- apply(mat, 1, function(x) {
    x <- x[!is.na(x)]
    length(x) > 1 && var(x) > 0
  })
  mat[keep, , drop = FALSE]
}
```

```r
# ============================================
# LOAD DATA
# ============================================

path_abun <- read.delim(
  "/Users/LuicerAnneIngasia/Desktop/wcs/humman/humann3_pathabundance_merged_normalised_UG.tsv",
  sep = "\t",
  check.names = FALSE
)

path_cov <- read.delim(
  "/Users/LuicerAnneIngasia/Desktop/wcs/humman/humann3_pathcoverage_merged_UG.tsv",
  sep = "\t",
  check.names = FALSE
)

genefam <- read.delim(
  "/Users/LuicerAnneIngasia/Desktop/wcs/humman/humann3_genefamilies_merged_normalised_UG.tsv",
  sep = "\t",
  check.names = FALSE
)

meta <- read.csv(
  "/Users/LuicerAnneIngasia/Desktop/wcs/Bednarski_2025_cleaned_metadata_final.csv",
  check.names = FALSE
)

dim(path_abun)
```

```
## [1] 22065    38
```

```r
dim(path_cov)
```

```
## [1] 22069    38
```

```r
dim(genefam)
```

```
## [1] 3278640      38
```

```r
dim(meta)
```

```
## [1] 39 68
```

```r
head(meta)
```

```
##         sample sample_name   metagm_id country studyid uric_se1 group acidotic
## 1 CM_1_met_abs    CM_1_S71 SRR30598656  Uganda    1109      1.8     1        0
## 2 CM_2_met_abs    CM_2_S72 SRR30598652  Uganda    1199      3.7     1        0
## 3 CM_3_met_abs    CM_3_S73 SRR30598651  Uganda    1384      6.8     1        0
## 4 CM_4_met_abs    CM_4_S74 SRR30598648  Uganda    1389      8.5     1        0
## 5 CM_5_met_abs    CM_5_S75 SRR30598640  Uganda    1450      3.0     1        0
## 6 CM_6_met_abs    CM_6_S76 SRR30598628  Uganda    1600      5.7     1        0
##   ceftri ceftri1 ceftri2 antibiot2 antibiot1 antibio2 chloram cipro xpen gentam
## 1      0       1       0         0         1        0       0     0    0      0
## 2      0       1       0         0         1        0       0     0    0      0
## 3      0       1       0         0         1        0       0     0    0      0
## 4      0       1       0         1         1        0       0     0    0      0
## 5      0       1       0         1         1        0       0     0    0      0
## 6      0       1       0         0         1        0       0     0    0      0
##   flagyl cotrimoxazole        doe  stool_dt0 chloram1 cipro1 xpen1 gentam1
## 1      0             0  3/24/2015  3/28/2015        0      0     0       0
## 2      0             0  3/22/2016  3/24/2016        0      0     0       1
## 3      0             0 03/03/2016 03/07/2016        0      0     0       0
## 4      0             0  2/24/2016  2/25/2016        0      0     0       0
## 5      0             0  9/17/2016  9/21/2016        0      0     0       0
## 6      0             0  1/26/2017  1/26/2017        0      0     0       0
##   flagyl1 amoxyl lacidosis tff3_pl1 outcome sm_death12m site sex  age hbsgt
## 1       0      0         0    1.259       0           0    1   1 3.46     1
## 2       0      0         0    1.774       0           0    2   1 3.97     1
## 3       0      0         0    2.184       0           0    1   1 3.67     1
## 4       0      0         0    1.849       0           0    1   2 3.86     1
## 5       0      0         0    1.674       0           0    1   1 3.87     1
## 6       0      0         0    2.143       0           0    1   2 3.09     1
##   haki inpatmal12m inpatmalno12m inpatany12m enr_sma hgbbase feedhour neuabs
## 1    0           0             0           0       0     8.5       48   4.26
## 2    0           1             1           1       0     9.6       53   4.17
## 3    0           1             1           1       0     7.4       32   5.57
## 4    0           0             0           0       0     7.6       24   9.61
## 5    0           0             0           0       0     7.3       18   8.46
## 6    0          NA            NA          NA       0     7.3       24   3.24
##    ho1_pl1 blcx_pos   wbc        dod trem_pl1 inpatsepsis12m deathdaysto12m
## 1  61.5300        0  6.93  3/30/2015   278.35              0             NA
## 2  58.8200        0  5.84 04/05/2016   425.56              0             NA
## 3 113.8275        0  6.79 03/09/2016   356.35              0             NA
## 4  73.6125        0 15.87  2/28/2016   265.79              0             NA
## 5 102.0850        0 13.22  9/22/2016   197.67              0             NA
## 6   7.9700        0  6.26  1/31/2017   203.40             NA             NA
##   blcx_result blcx_result_spec othantibi  whz0  haz0  waz0 pcr_fal0   doe_date
## 1          NA               NA        NA -0.78 -2.54 -1.96        1 2015-03-24
## 2          NA               NA        NA -2.03 -0.29 -1.49        1 2016-03-22
## 3          NA               NA        NA  0.71 -2.58 -1.03        1 2016-03-03
## 4           6               NA        NA -1.18 -1.09 -1.45        1 2016-02-24
```

```
## 5          NA               NA        NA -0.99 -2.91 -2.35        1 2016-09-17
## 6          NA               NA        NA -1.38 -1.16 -1.59        1 2017-01-26
##   admission_month    Sex   Site hbsgt_bin age_in_months prior_abx_oral
## 1               3   Male Mulago     AA/AS            42              0
## 2               3   Male  Jinja     AA/AS            48              0
## 3               3   Male Mulago     AA/AS            44              0
## 4               2 Female Mulago     AA/AS            46              0
## 5               9   Male Mulago     AA/AS            46              0
## 6               1 Female Mulago     AA/AS            37              0
##   date_to_stool stool_day sick
## 1             4     day 4   SM
## 2             2     day 2   SM
## 3             4     day 4   SM
## 4             1     day 1   SM
## 5             4     day 4   SM
## 6             0     day 0   SM
```

```r
# ============================================
# CLEAN DATA
# ============================================

colnames(path_abun)[1] <- "Feature"
colnames(path_cov)[1]  <- "Feature"
colnames(genefam)[1]   <- "Feature"

path_abun <- path_abun %>%
  filter(!grepl("UNMAPPED|UNINTEGRATED", Feature))

path_cov <- path_cov %>%
  filter(!grepl("UNMAPPED|UNINTEGRATED", Feature))

genefam <- genefam %>%
  filter(!grepl("UNMAPPED|UNINTEGRATED", Feature))

# Keep unstratified rows for main functional analysis
path_abun_unstrat <- path_abun %>%
  filter(!grepl("\\|", Feature))

path_cov_unstrat <- path_cov %>%
  filter(!grepl("\\|", Feature))
```

```r
# ============================================
# CHECK COLUMN NAMING
# ============================================

head(colnames(path_abun_unstrat))
```

```
## [1] "Feature"               "SRR30598618_Abundance" "SRR30598619_Abundance"
## [4] "SRR30598620_Abundance" "SRR30598621_Abundance" "SRR30598622_Abundance"
```

```r
head(colnames(path_cov_unstrat))
```

```
## [1] "Feature"               "SRR30598618_Coverage" "SRR30598619_Coverage"
## [4] "SRR30598620_Coverage" "SRR30598621_Coverage" "SRR30598622_Coverage"
```

```r
head(colnames(genefam))
```

```
## [1] "Feature"                     "SRR30598618_Abundance-RPKs"
## [3] "SRR30598619_Abundance-RPKs" "SRR30598620_Abundance-RPKs"
## [5] "SRR30598621_Abundance-RPKs" "SRR30598622_Abundance-RPKs"
```

```r
ncol(path_abun_unstrat)
```

```
## [1] 38
```

```r
ncol(path_cov_unstrat)
```

```
## [1] 38
```

```r
ncol(genefam)
```

```
## [1] 38
```

```r
# ============================================
# MATCH METADATA TO HUMAnN SAMPLES
# ============================================

meta2 <- meta %>%
  filter(!is.na(metagm_id), metagm_id != "")

abun_ids <- extract_run_id(colnames(path_abun_unstrat)[-1])
cov_ids  <- extract_run_id(colnames(path_cov_unstrat)[-1])
gene_ids <- extract_run_id(colnames(genefam)[-1])

cat("Pathway abundance samples:", length(abun_ids), "\n")
```

```
## Pathway abundance samples: 37
```

```r
cat("Pathway coverage samples:", length(cov_ids), "\n")
```

```
## Pathway coverage samples: 37
```

```r
cat("Gene family samples:", length(gene_ids), "\n")
```

```
## Gene family samples: 37
```

```r
cat("Metadata samples with metagm_id:", nrow(meta2), "\n")
```

```
## Metadata samples with metagm_id: 39
```

```r
shared_ids <- Reduce(intersect, list(abun_ids, cov_ids, gene_ids, meta2$metagm_id))
shared_ids <- abun_ids[abun_ids %in% shared_ids]

cat("Shared samples across all data types and metadata:", length(shared_ids), "\n")
```

```
## Shared samples across all data types and metadata: 37
```

```r
abun_keep <- match(shared_ids, abun_ids)
cov_keep  <- match(shared_ids, cov_ids)
gene_keep <- match(shared_ids, gene_ids)

path_abun_unstrat <- path_abun_unstrat[, c(1, abun_keep + 1), drop = FALSE]
path_cov_unstrat  <- path_cov_unstrat[, c(1, cov_keep + 1), drop = FALSE]
genefam <- genefam[, c(1, gene_keep + 1), drop = FALSE]

meta2 <- meta2[match(shared_ids, meta2$metagm_id), , drop = FALSE]

stopifnot(all(extract_run_id(colnames(path_abun_unstrat)[-1]) == shared_ids))
stopifnot(all(extract_run_id(colnames(path_cov_unstrat)[-1]) == shared_ids))
stopifnot(all(extract_run_id(colnames(genefam)[-1]) == shared_ids))
stopifnot(all(meta2$run_accession == shared_ids))

meta <- meta2
meta$sick <- as.factor(meta$sick)

colnames(path_abun_unstrat)[-1] <- meta$sample_name
colnames(path_cov_unstrat)[-1]  <- meta$sample_name
colnames(genefam)[-1]           <- meta$sample_name

cat("Final matched sample count:", nrow(meta), "\n")
```

```
## Final matched sample count: 37
```

```r
table(meta$sick)
```

```
##
## CC SM
##  9 28
```

```r
head(meta[, c("sample_name", "metagm_id", "sick", "country")])
```

```
##     sample_name   metagm_id sick country
## 28    SMA_1_S80 SRR30598618   SM  Uganda
## 36    SMA_6_S85 SRR30598619   SM  Uganda
## 17 pfpos_8_S109 SRR30598620   CC  Uganda
## 26   RDS_8_S100 SRR30598621   SM  Uganda
## 32    SMA_2_S81 SRR30598622   SM  Uganda
## 29   SMA_10_S89 SRR30598623   SM  Uganda
```

Question 1: What are the most abundant microbial functions across the cohort?

These first plots are exploratory. They help us understand broad functional structure before testing specific associations.

```
# ============================================
# PATHWAY MATRIX
# ============================================

path_mat <- path_abun_unstrat %>%
  column_to_rownames("Feature") %>%
  as.matrix()

mode(path_mat) <- "numeric"
path_mat_log <- log10(path_mat + 1e-6)
```

We begin by examining global structure using dimensionality reduction. Principal Component Analysis (PCA) allows us to assess whether samples cluster by group.

```
# ============================================
# PCA OF PATHWAY ABUNDANCE
# ============================================

pca_input <- t(path_mat_log)

keep_cols <- apply(
  pca_input,
  2,
  function(x) {
    x <- x[!is.na(x)]
    length(x) > 1 && var(x) > 0
  }
)

pca_input <- pca_input[, keep_cols, drop = FALSE]

cat("Samples used for PCA:", nrow(pca_input), "\n")
```

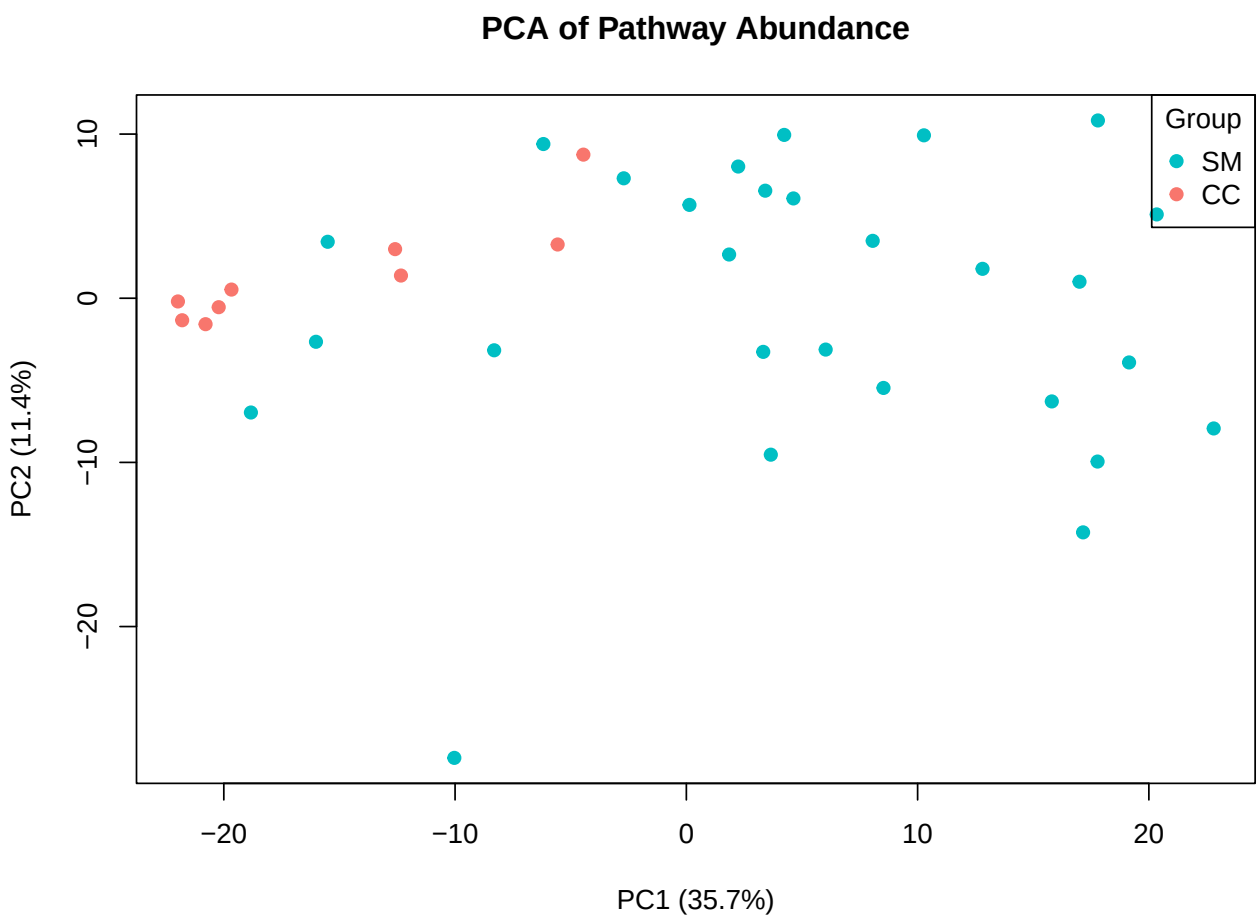
```
## Samples used for PCA: 37
```

```
cat("Pathways retained for PCA:", ncol(pca_input), "\n")
```

```
## Pathways retained for PCA: 537
```

```
pca <- prcomp(pca_input, scale. = TRUE)

meta <- meta[match(rownames(pca$x), meta$sample_name), ]
meta$sick <- factor(meta$sick, levels = c("SM", "CC"))

{group_colors <- c("SM" = "#00BFC4", "CC" = "#F8766D")
point_colors <- group_colors[meta$sick]
```

```r
plot(
  pca$x[, 1], pca$x[, 2],
  col = point_colors,
  pch = 19,
  xlab = paste0("PC1 (", round(100 * summary(pca)$importance[2, 1], 1), "%)"),
  ylab = paste0("PC2 (", round(100 * summary(pca)$importance[2, 2], 1), "%)"),
  main = "PCA of Pathway Abundance"
)

legend(
  "topright",
  legend = names(group_colors),
  col = group_colors,
  pch = 19,
  title = "Group"
)}
```



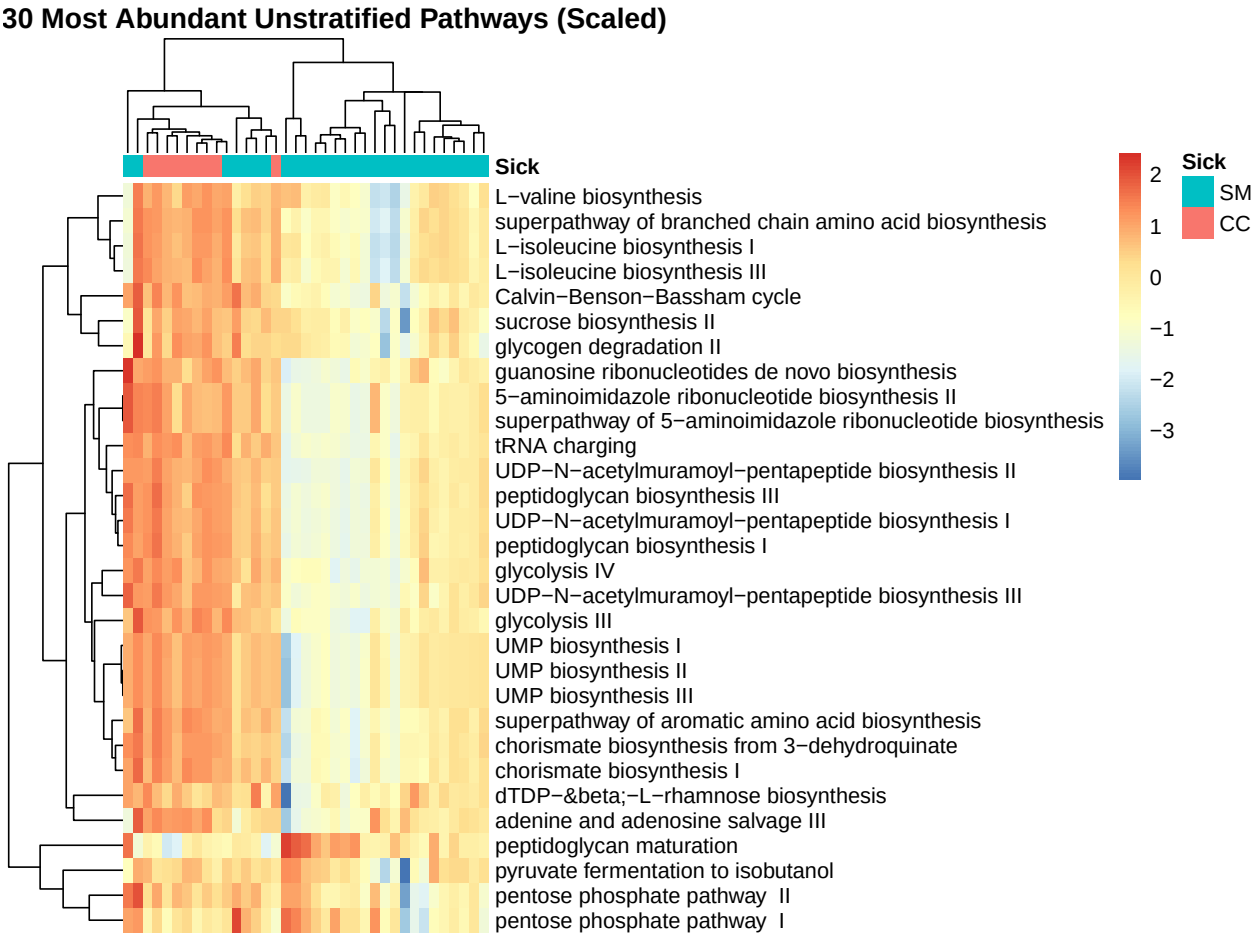

```r
table(meta$sick)
```

```
##
## SM CC
## 28  9
```

Interpretation prompt:

Do samples separate clearly by group, or is the clustering partial? The PCA plot shows partial separation by outcome along PC1 (35.7% variance explained), with some clustering of the CC group, but there is substantial overlap between groups. PC2 (11.4%) captures additional variability but does not clearly distinguish groups. Incomplete separation suggests the functional shifts are subtle rather than absolute.

```r
# ============================================
# TOP ABUNDANT PATHWAYS
# ============================================

path_means <- rowMeans(path_mat, na.rm = TRUE)
top_pathways <- names(sort(path_means, decreasing = TRUE))[1:30]

heat_path <- path_mat_log[top_pathways, , drop = FALSE]
heat_path_scaled <- t(scale(t(heat_path)))

rownames(heat_path_scaled) <- clean_names(rownames(heat_path_scaled))

# Optional: colors for annotation
ann_colors <- list(
  Sick = c(
    SM = "#00BFC4",
    CC    = "#F8766D"
  )
)

# Create annotation for columns (samples)
annotation_col <- data.frame(
  Sick = meta$sick
)
rownames(annotation_col) <- meta$sample_name  # must match target_log columns

pheatmap(
  heat_path_scaled,
  show_rownames = TRUE,
  show_colnames = FALSE,
  main = "Top 30 Most Abundant Unstratified Pathways (Scaled)",
  annotation_col = annotation_col,
  annotation_colors = ann_colors
)
```



Interpretation prompt:
**These pathways usually reflect core metabolism and microbial growth rather than group-specific biology*. The heatmap of the top 30 most abundant pathways shows that microbial functional profiles are dominated by core metabolic processes, including amino acid biosynthesis (e.g., branched-chain amino acids), nucleotide biosynthesis, glycolysis, and cell wall (peptidoglycan) synthesis. These pathways are consistently abundant across samples, reflecting essential functions required for microbial growth and maintenance.

While there is variation in intensity across samples, these differences are not strictly group-specific, indicating that these core pathways are broadly shared across both groups. Some samples in the sick group (CC) show localized increases or decreases in specific pathways, but these patterns are not uniformly consistent across all individuals.

This suggests that while there are some group-associated differences in pathway abundance, the overall functional landscape of the microbiome is largely conserved, and differences between groups are subtle rather than driven by major shifts in dominant functions.

Question 2: How consistently are pathways detected across samples?

```
# ==========================================
# PATHWAY COVERAGE
# ==========================================

cov_mat <- path_cov_unstrat %>%
```

```r
  column_to_rownames("Feature") %>%
  as.matrix()

mode(cov_mat) <- "numeric"

prev <- rowSums(cov_mat > 0, na.rm = TRUE)
cov_filtered <- cov_mat[prev > 0.2 * ncol(cov_mat), , drop = FALSE]

coverage_summary <- data.frame(
  Feature = rownames(cov_mat),
  Prevalence = prev,
  Proportion = prev / ncol(cov_mat)
) %>%
  arrange(desc(Proportion))

dim(cov_filtered)
```

```
## [1] 126  37
```

```r
head(coverage_summary, 20)
```

```
##
## GLUCOSE1PMETAB-PWY: glucose and glucose-1-phosphate degradation
## ILEUSYN-PWY: L-isoleucine biosynthesis I (from threonine)
## PEPTIDOGLYCANSYN-PWY: peptidoglycan biosynthesis I (meso-diaminopimelate containing)
## PWY-5103: L-isoleucine biosynthesis III
## PWY-5667: CDP-diacylglycerol biosynthesis I
## PWY-5695: inosine 5'-phosphate degradation
## PWY-5941: glycogen degradation II
## PWY-6121: 5-aminoimidazole ribonucleotide biosynthesis I
## PWY-6122: 5-aminoimidazole ribonucleotide biosynthesis II
## PWY-6277: superpathway of 5-aminoimidazole ribonucleotide biosynthesis
## PWY-6385: peptidoglycan biosynthesis III (mycobacteria)
## PWY-6386: UDP-N-acetylmuramoyl-pentapeptide biosynthesis II (lysine-containing)
## PWY-6387: UDP-N-acetylmuramoyl-pentapeptide biosynthesis I (meso-diaminopimelate containing)     PWY-
## PWY-6609: adenine and adenosine salvage III
## PWY-7221: guanosine ribonucleotides de novo biosynthesis
## PWY-7663: gondoate biosynthesis (anaerobic)
## PWY-7953: UDP-N-acetylmuramoyl-pentapeptide biosynthesis III (meso-diaminopimelate containing) PWY-79
## PWY-7977: L-methionine biosynthesis IV
## PWY0-1319: CDP-diacylglycerol biosynthesis II
## PWY0-1586: peptidoglycan maturation (meso-diaminopimelate containing)
##                                                                                                Preval
## GLUCOSE1PMETAB-PWY: glucose and glucose-1-phosphate degradation
## ILEUSYN-PWY: L-isoleucine biosynthesis I (from threonine)
## PEPTIDOGLYCANSYN-PWY: peptidoglycan biosynthesis I (meso-diaminopimelate containing)
## PWY-5103: L-isoleucine biosynthesis III
## PWY-5667: CDP-diacylglycerol biosynthesis I
## PWY-5695: inosine 5'-phosphate degradation
## PWY-5941: glycogen degradation II
## PWY-6121: 5-aminoimidazole ribonucleotide biosynthesis I
## PWY-6122: 5-aminoimidazole ribonucleotide biosynthesis II
```

```
## PWY-6277: superpathway of 5-aminoimidazole ribonucleotide biosynthesis
## PWY-6385: peptidoglycan biosynthesis III (mycobacteria)
## PWY-6386: UDP-N-acetylmuramoyl-pentapeptide biosynthesis II (lysine-containing)
## PWY-6387: UDP-N-acetylmuramoyl-pentapeptide biosynthesis I (meso-diaminopimelate containing)
## PWY-6609: adenine and adenosine salvage III
## PWY-7221: guanosine ribonucleotides de novo biosynthesis
## PWY-7663: gondoate biosynthesis (anaerobic)
## PWY-7953: UDP-N-acetylmuramoyl-pentapeptide biosynthesis III (meso-diaminopimelate containing)
## PWY-7977: L-methionine biosynthesis IV
## PWY0-1319: CDP-diacylglycerol biosynthesis II
## PWY0-1586: peptidoglycan maturation (meso-diaminopimelate containing)
##                                                                                                  Propo
## GLUCOSE1PMETAB-PWY: glucose and glucose-1-phosphate degradation
## ILEUSYN-PWY: L-isoleucine biosynthesis I (from threonine)
## PEPTIDOGLYCANSYN-PWY: peptidoglycan biosynthesis I (meso-diaminopimelate containing)
## PWY-5103: L-isoleucine biosynthesis III
## PWY-5667: CDP-diacylglycerol biosynthesis I
## PWY-5695: inosine 5'-phosphate degradation
## PWY-5941: glycogen degradation II
## PWY-6121: 5-aminoimidazole ribonucleotide biosynthesis I
## PWY-6122: 5-aminoimidazole ribonucleotide biosynthesis II
## PWY-6277: superpathway of 5-aminoimidazole ribonucleotide biosynthesis
## PWY-6385: peptidoglycan biosynthesis III (mycobacteria)
## PWY-6386: UDP-N-acetylmuramoyl-pentapeptide biosynthesis II (lysine-containing)
## PWY-6387: UDP-N-acetylmuramoyl-pentapeptide biosynthesis I (meso-diaminopimelate containing)
## PWY-6609: adenine and adenosine salvage III
## PWY-7221: guanosine ribonucleotides de novo biosynthesis
## PWY-7663: gondoate biosynthesis (anaerobic)
## PWY-7953: UDP-N-acetylmuramoyl-pentapeptide biosynthesis III (meso-diaminopimelate containing)
## PWY-7977: L-methionine biosynthesis IV
## PWY0-1319: CDP-diacylglycerol biosynthesis II
## PWY0-1586: peptidoglycan maturation (meso-diaminopimelate containing)
```

Interpretation prompt:
Pathway abundance asks how much of a pathway is present. Pathway coverage asks how consistently a pathway is represented across samples.

Pathway coverage analysis shows that a subset of pathways is consistently detected across a large proportion of samples, indicating the presence of a shared core functional repertoire. These highly prevalent pathways likely correspond to essential microbial functions that are maintained across individuals.

At the same time, many pathways show lower prevalence and are only detected in a subset of samples, reflecting functional variability within the cohort. This suggests that while core metabolic functions are stable, additional pathways may be condition-specific or influenced by host or environmental factors.

Importantly, pathway coverage complements pathway abundance: abundance reflects how much of a function is present, whereas coverage indicates how consistently that function is observed across samples.

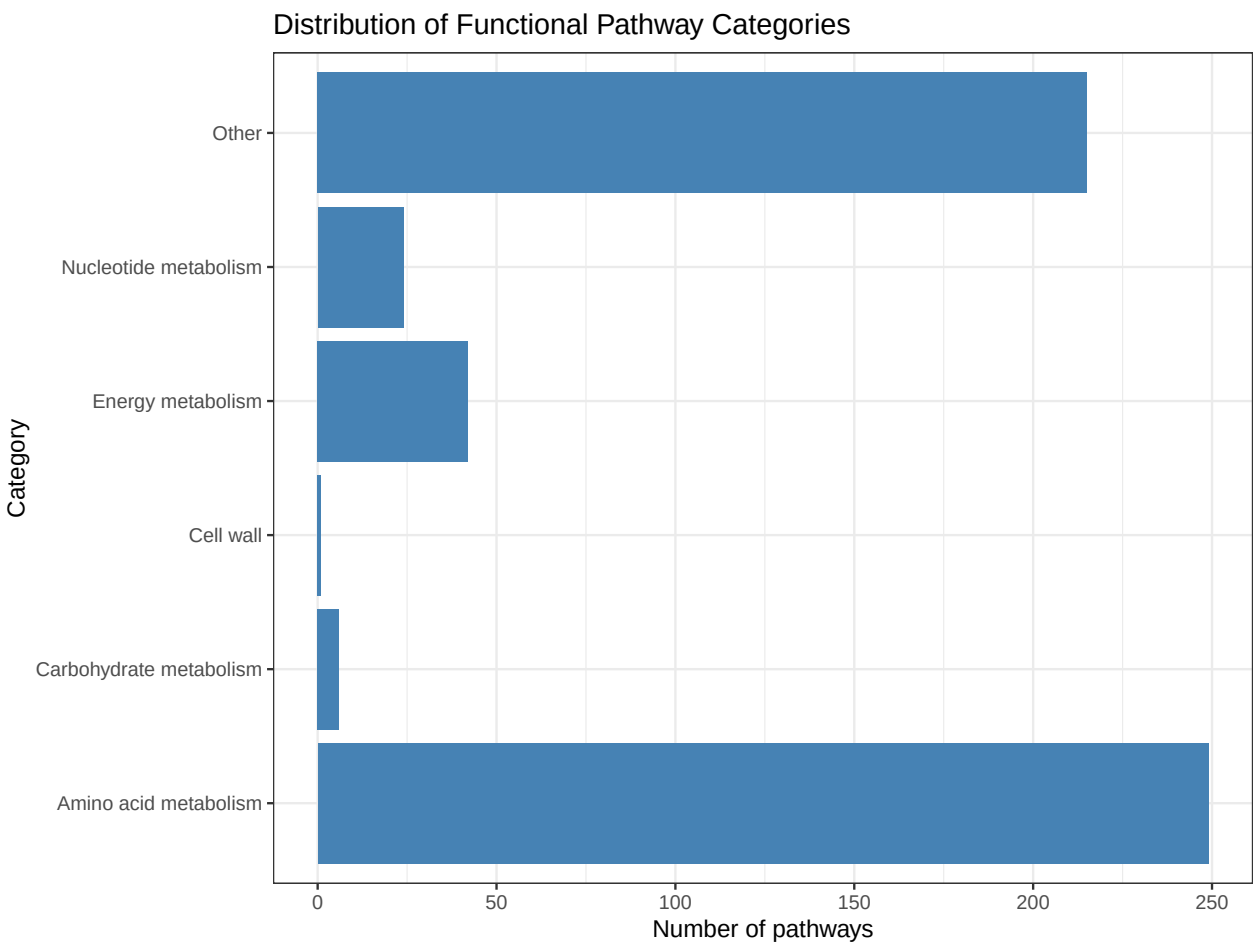
Question 3: Which broad biological systems dominate the microbiome?

```
# ============================================
# FUNCTIONAL CATEGORIES
# ============================================
```

```r
pathway_names <- clean_names(rownames(path_mat))
categories <- annotate_pathways(pathway_names)

table(categories)
```

```
## categories
##   Amino acid metabolism Carbohydrate metabolism                Cell wall
##                     249                       6                        1
##       Energy metabolism   Nucleotide metabolism                    Other
##                      42                      24                      215
```

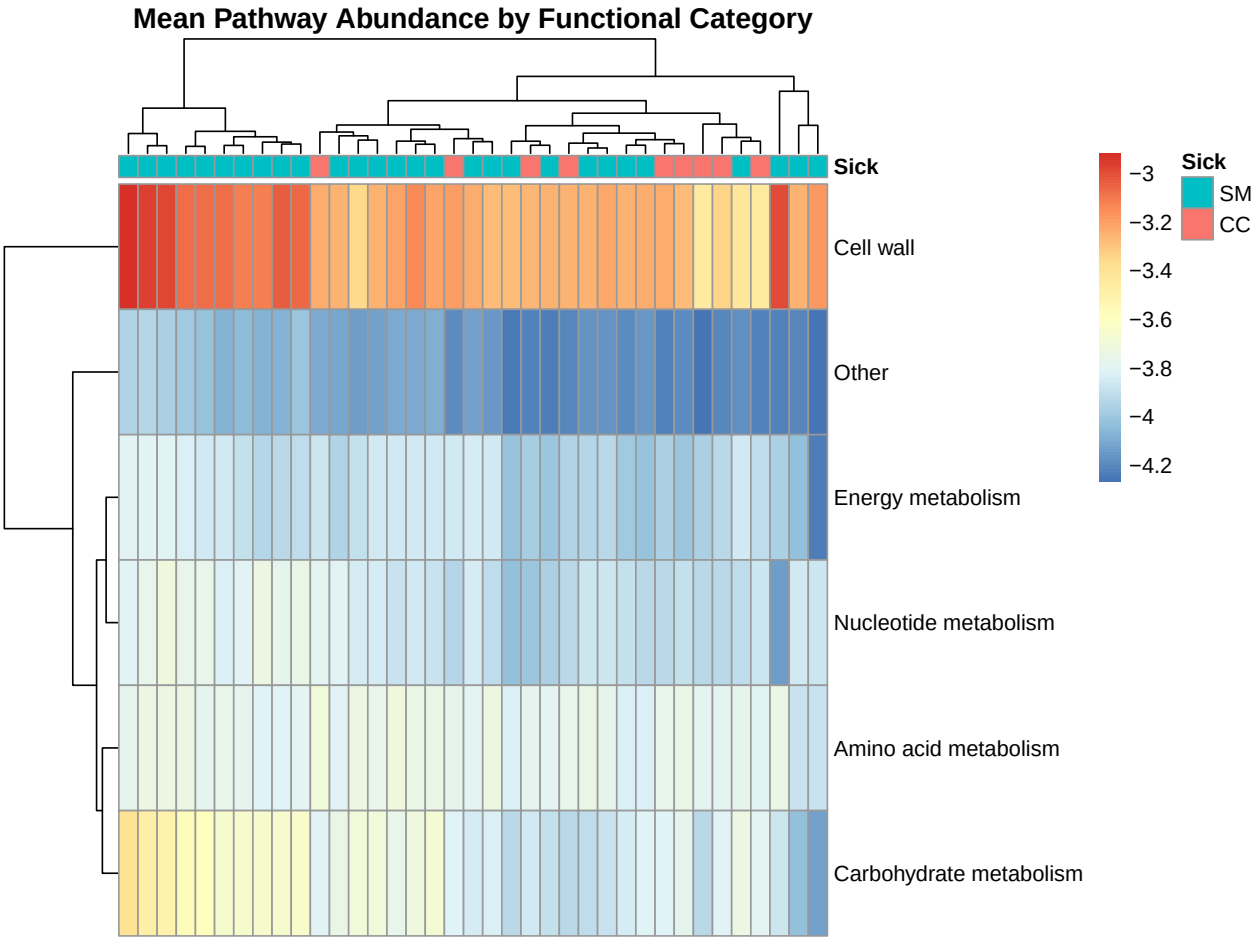
```r
cat_df <- data.frame(category = categories)

ggplot(cat_df, aes(x = category)) +
  geom_bar(fill = "steelblue") +
  theme_bw() +
  coord_flip() +
  labs(
    title = "Distribution of Functional Pathway Categories",
    x = "Category",
    y = "Number of pathways"
  )
```



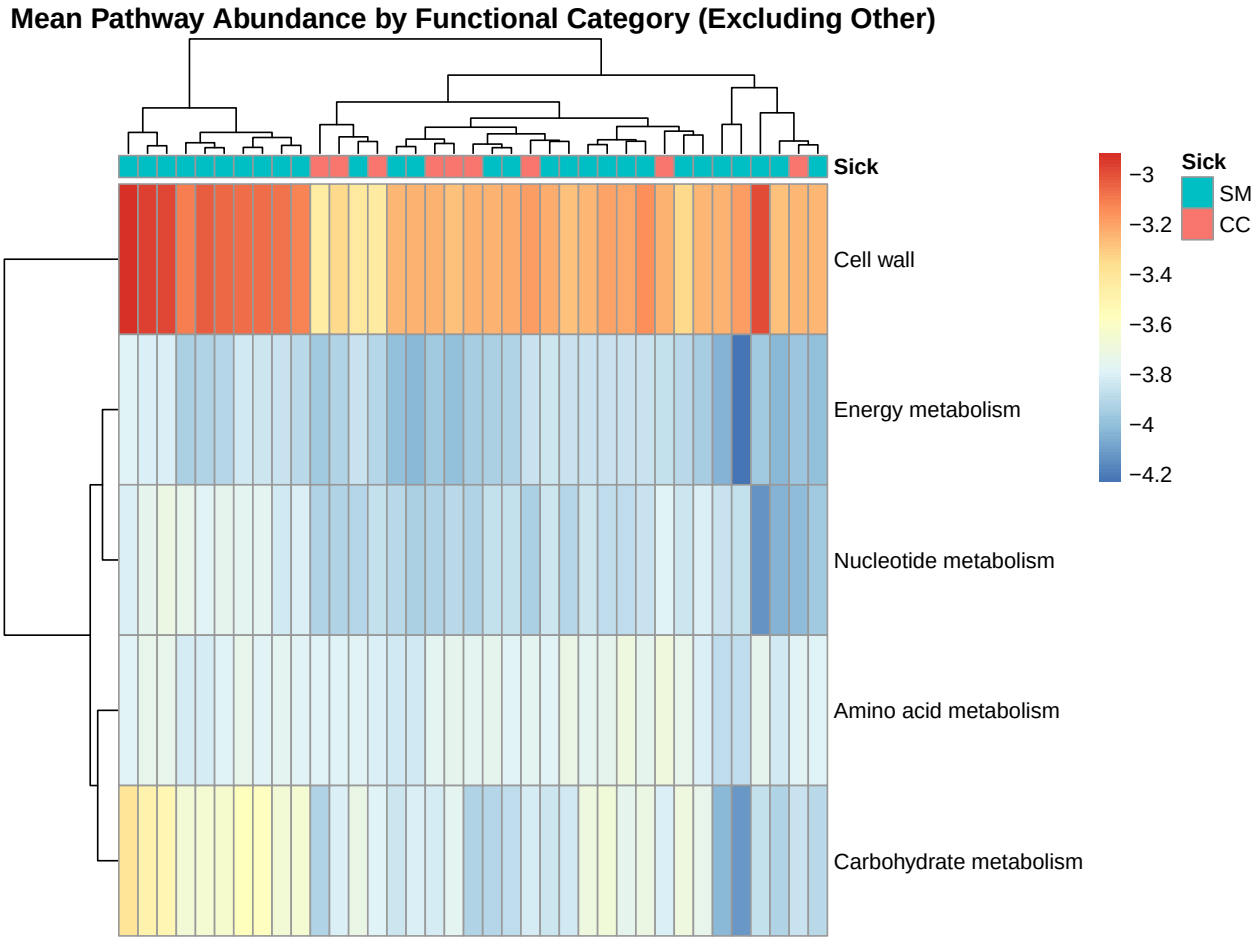
Interpretation prompt:
This plot shows how many pathways fall into each broad biological category. It helps move from pathway IDs to interpretable biology.

Functional categorisation shows that the microbiome is dominated by pathways related to metabolism, particularly energy metabolism, carbohydrate metabolism, and amino acid metabolism. These categories contain the largest number of pathways, as seen in the category distribution plot, indicating that microbial communities are primarily structured around nutrient processing and energy production.

```r
# ============================================
# MEAN ABUNDANCE BY FUNCTIONAL CATEGORY
# ============================================

path_df <- as.data.frame(path_mat)
path_df$category <- categories

category_abundance_mean <- path_df %>%
  group_by(category) %>%
  summarise(across(where(is.numeric), mean))

cat_mat_mean <- category_abundance_mean %>%
  column_to_rownames("category") %>%
  as.matrix()

cat_mat_mean_log <- log10(cat_mat_mean + 1e-6)

pheatmap(
  cat_mat_mean_log,
  main = "Mean Pathway Abundance by Functional Category",
  cluster_rows = TRUE,
  cluster_cols = TRUE,
  show_colnames = FALSE,
  annotation_col = annotation_col,
  annotation_colors = ann_colors
)
```




```r
# ==========================================
# EXCLUDING THE BROAD 'OTHER' CATEGORY
# ==========================================

cat_mat_mean_no_other <- cat_mat_mean[rownames(cat_mat_mean) != "Other", , drop = FALSE]

pheatmap(
  log10(cat_mat_mean_no_other + 1e-6),
  main = "Mean Pathway Abundance by Functional Category (Excluding Other)",
  cluster_rows = TRUE,
  cluster_cols = TRUE,
  show_colnames = FALSE,
  annotation_col = annotation_col,
  annotation_colors = ann_colors
)
```



Interpretation prompt:
This aggregation is more interpretable than individual features and makes it easier to discuss pathways in terms of metabolism, cell wall processes, and nucleotide or energy functions.

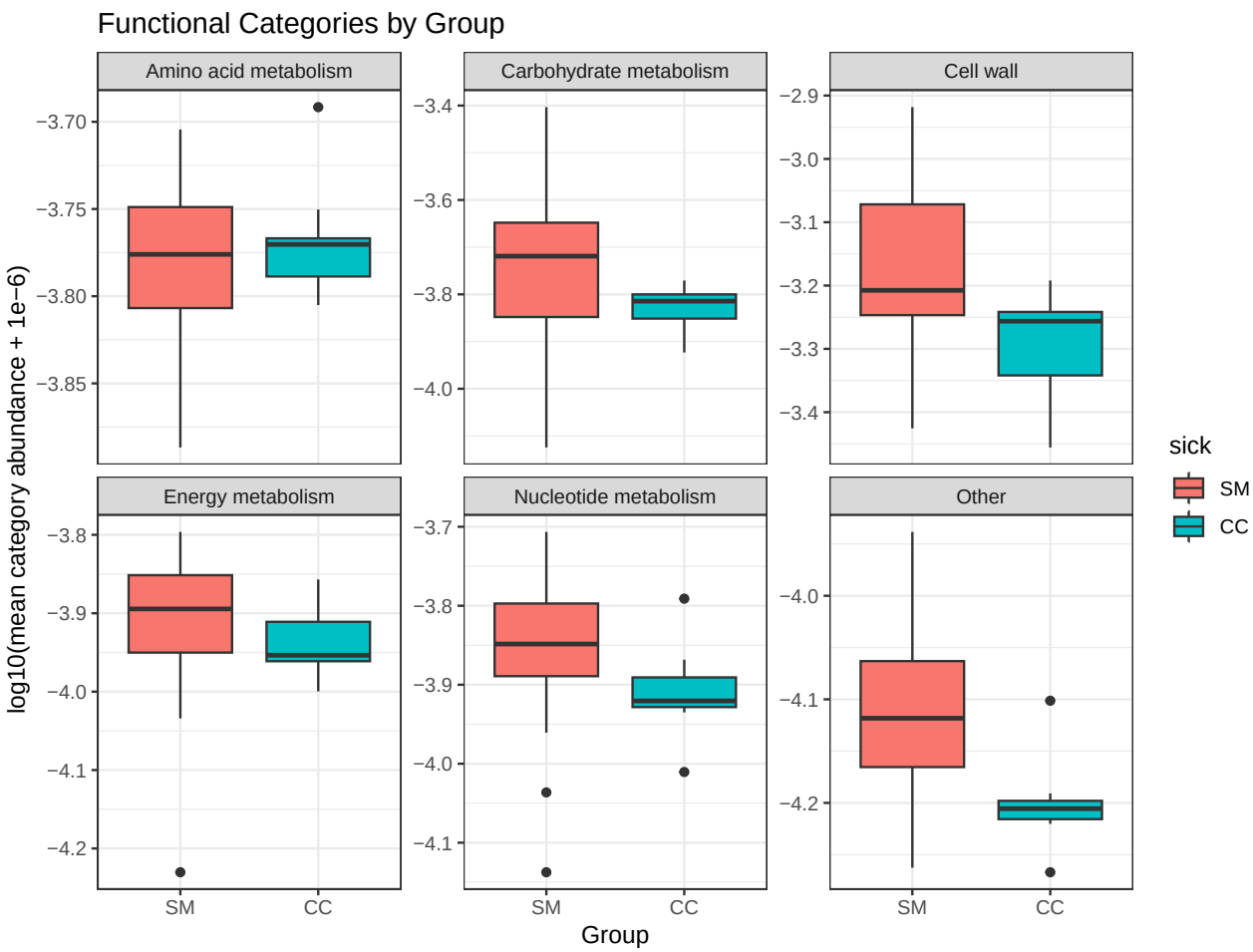
The heatmap of mean category abundance further supports this, showing that these metabolic categories consistently exhibit higher abundance across samples compared to other functional groups. In contrast, categories such as cell wall processes and cofactor-related pathways are present but contribute less overall to the functional landscape.

When the broad "Other" category is excluded, the remaining categories become more interpretable, highlighting that core metabolic processes are the dominant functional components of the microbiome.

Question 4: Do broad biological categories differ across groups?

```
# ============================================
# CATEGORY ABUNDANCE BY GROUP
# ============================================

cat_df_group <- as.data.frame(t(cat_mat_mean))
cat_df_group$sick <- meta$sick

cat_long <- cat_df_group %>%
  pivot_longer(-sick, names_to = "category", values_to = "abundance")
```

```r
ggplot(cat_long, aes(x = sick, y = log10(abundance + 1e-6), fill = sick)) +
  geom_boxplot() +
  facet_wrap(~category, scales = "free_y") +
  theme_bw() +
  labs(
    title = "Functional Categories by Group",
    x = "Group",
    y = "log10(mean category abundance + 1e-6)"
  )
```



Interpretation prompt:
Do certain groups show relatively higher energy metabolism, nucleotide metabolism, or cell-wall-associated pathways? These plots summarize broad biology before moving into feature-level testing.

Boxplots comparing functional categories across groups show that differences between SM and CC are generally modest and category-specific rather than global. Most categories exhibit substantial overlap between groups, indicating that overall functional composition remains broadly similar.

Some categories, particularly those related to metabolism (e.g., energy, carbohydrate, or amino acid metabolism), show slight shifts in distribution between groups. However, these differences are not consistent across all samples and do not indicate a uniform increase or decrease within a group.

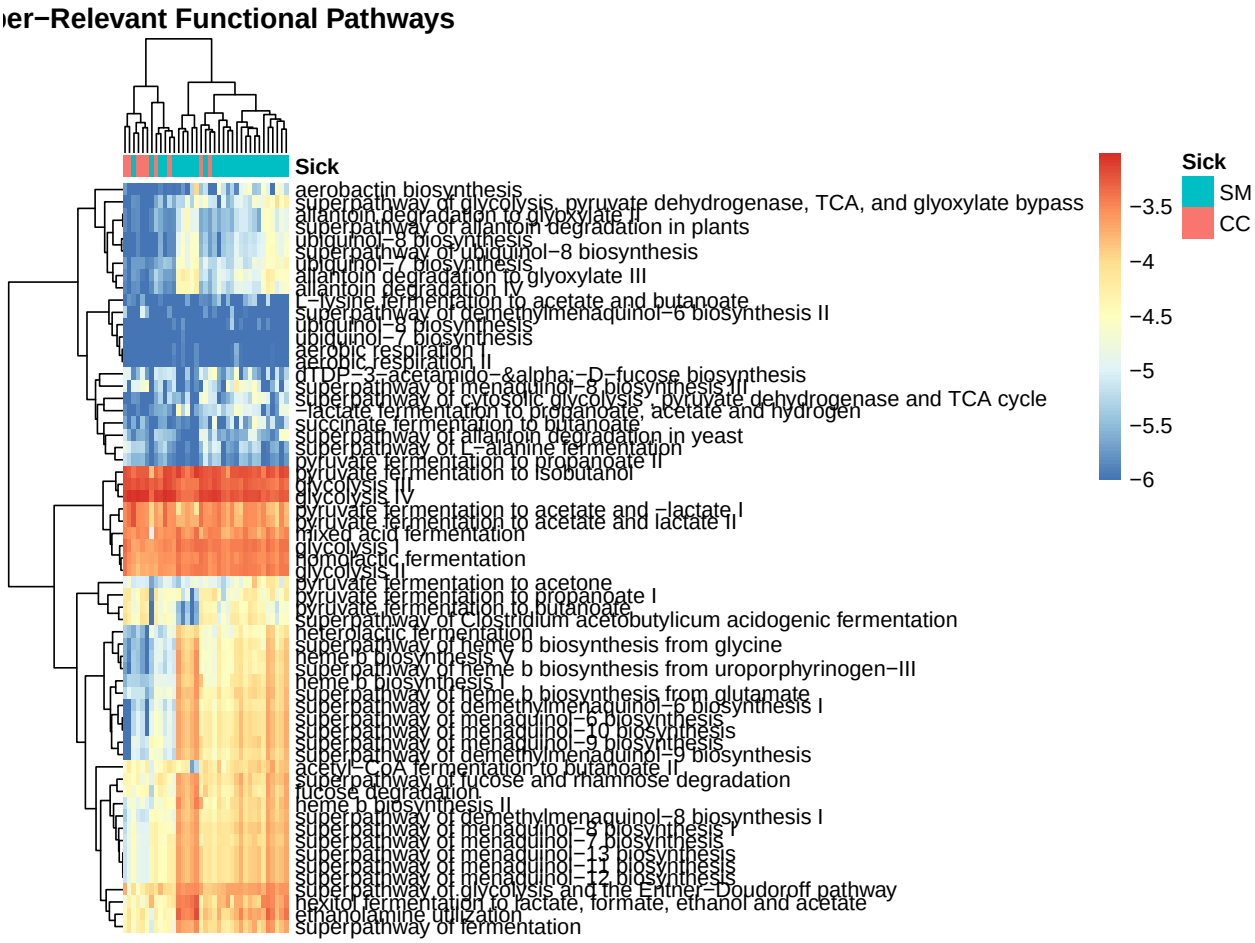
Because each category is plotted on a different y-axis scale, the apparent differences should be interpreted cautiously. The results suggest that group-associated functional changes are subtle and likely confined to specific pathways rather than broad functional classes.

These patterns indicate that functional differences between groups are likely driven by specific pathways rather than broad shifts in entire biological systems.

Question 5: Do we see paper-relevant functional themes?

The paper highlights metabolism, respiration, iron/heme handling, and nutrient utilization. We can inspect those themes directly.

```
# ============================================
# PAPER-FOCUSED PATHWAYS
# ============================================

clean_features <- clean_names(rownames(path_mat))

target_idx <- grepl(
  "respiration|iron|heme|glycolysis|fermentation|aerobactin|fucose|sialic|allantoin|ethanolamine|ubiqui
  clean_features,
  ignore.case = TRUE
)

target_mat <- path_mat[target_idx, , drop = FALSE]
rownames(target_mat) <- clean_names(rownames(target_mat))
target_log <- log10(target_mat + 1e-6)


if (nrow(target_log) > 1) {
  pheatmap(
    target_log,
    show_rownames = TRUE,
    show_colnames = FALSE,
    main = "Paper-Relevant Functional Pathways",
    annotation_col = annotation_col,
    annotation_colors = ann_colors
  )
}
```



Interpretation prompt:
Are these functions consistently elevated in some groups? Do they suggest microbial communities adapted to altered nutrient availability, host-derived substrates, or stressed gut environments? Focusing on pathways highlighted in the paper—such as respiration, iron/heme acquisition, and host-derived nutrient utilisation—reveals targeted variation across samples and between groups.

The heatmap shows that these pathways are not uniformly distributed across all samples but instead exhibit sample-specific patterns of abundance. Certain pathways, particularly those related to respiration and iron/cofactor metabolism, display localized increases in subsets of samples, suggesting functional adaptation to specific host or environmental conditions.

Group-level patterns are present but not uniform: while some CC samples appear to show elevated abundance in pathways related to respiration or nutrient utilisation, there is substantial overlap with SM samples. This indicates that these functions are not exclusive to one group but may be enriched in certain individuals within a group.

Overall, these results suggest that biologically relevant functional shifts—particularly in energy metabolism, iron handling, and host-derived nutrient use—are present but are heterogeneous across the cohort. This supports the idea that microbiome adaptation occurs in a context-dependent manner rather than as a consistent group-wide signature.

These patterns are consistent with microbial communities responding to altered host environments, such as changes in oxygen availability, inflammation, or nutrient accessibility.

Question 6: What happens at the gene-family level?

Gene-family tables are much larger and more granular than pathways, so filtering is essential.

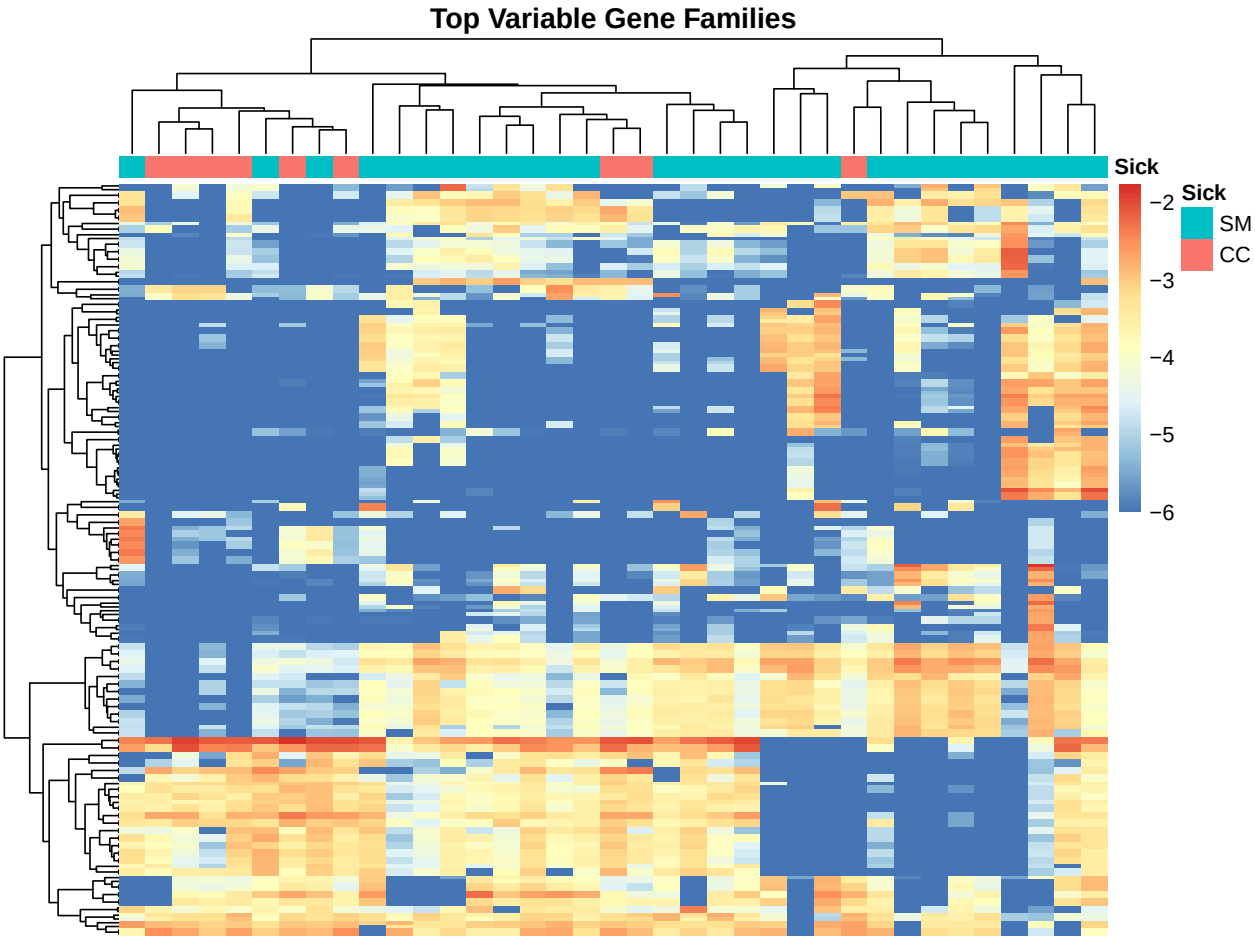
```r
# ============================================
# GENE FAMILY ANALYSIS
# ============================================

genefam_mat <- genefam %>%
  column_to_rownames("Feature") %>%
  as.matrix()

mode(genefam_mat) <- "numeric"

gene_prev <- rowMeans(genefam_mat > 0, na.rm = TRUE)
gene_mean <- rowMeans(genefam_mat, na.rm = TRUE)

genefam_filt <- genefam_mat[
  gene_prev >= 0.10 & gene_mean >= 1e-05,
  ,
  drop = FALSE
]

cat("Original gene families:", nrow(genefam_mat), "\n")
```

```
## Original gene families: 3278639
```

```r
cat("Retained after prevalence + mean-abundance filtering:", nrow(genefam_filt), "\n")
```

```
## Retained after prevalence + mean-abundance filtering: 16267
```

```r
# ============================================
# TOP VARIABLE FILTERED GENE FAMILIES
# ============================================

var_genes <- apply(genefam_filt, 1, var, na.rm = TRUE)

n_keep <- min(200, length(var_genes))
top_gene_names <- names(sort(var_genes, decreasing = TRUE))[1:n_keep]

genefam_top <- genefam_filt[top_gene_names, , drop = FALSE]
genefam_log <- log10(genefam_top + 1e-6)

pheatmap(
  genefam_log,
  show_rownames = FALSE,
  show_colnames = FALSE,
  main = "Top Variable Gene Families",
  annotation_col = annotation_col,
  annotation_colors = ann_colors
)
```



Interpretation prompt:
Gene families offer higher resolution, but they are harder to interpret biologically without annotation or targeted grouping.

Gene-family analysis reveals a much higher level of complexity compared to pathway-level results, reflecting the fine-grained nature of these features. After applying prevalence and abundance filtering, a reduced subset of gene families was retained for analysis, indicating that a large proportion of features are either rare or low-abundance.

The heatmap of the most variable gene families shows heterogeneous patterns across samples, with no clear or consistent clustering by group. Instead, variation appears to be driven primarily by individual samples rather than group-level structure.

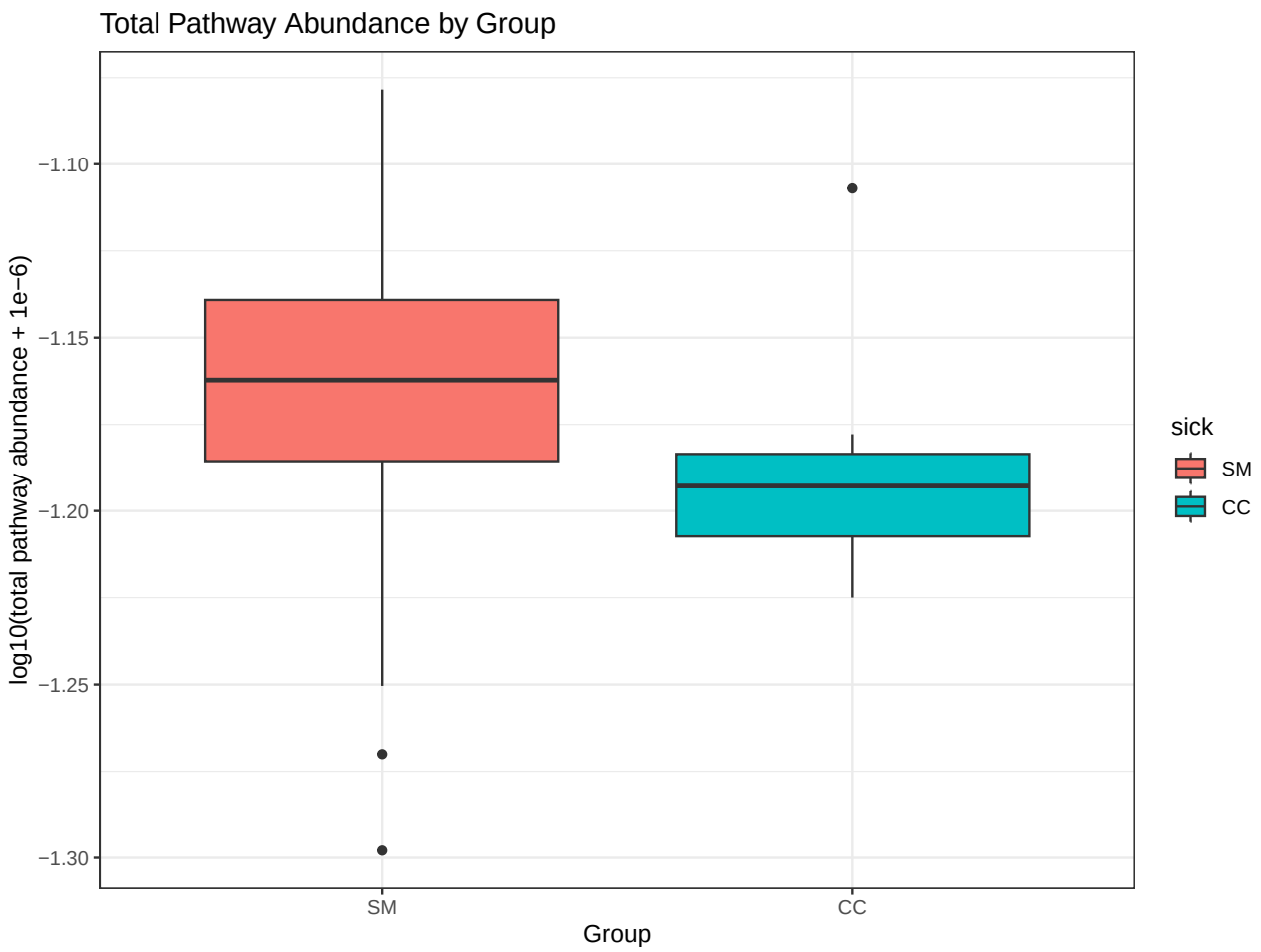
This contrasts with pathway-level analysis, where broader functional trends were more apparent. The gene-family level captures detailed functional potential, but this increased resolution comes with increased noise and reduced interpretability.

Overall, these results highlight that gene-family data provide high-resolution insights but require aggregation or targeted grouping to extract clear biological meaning.

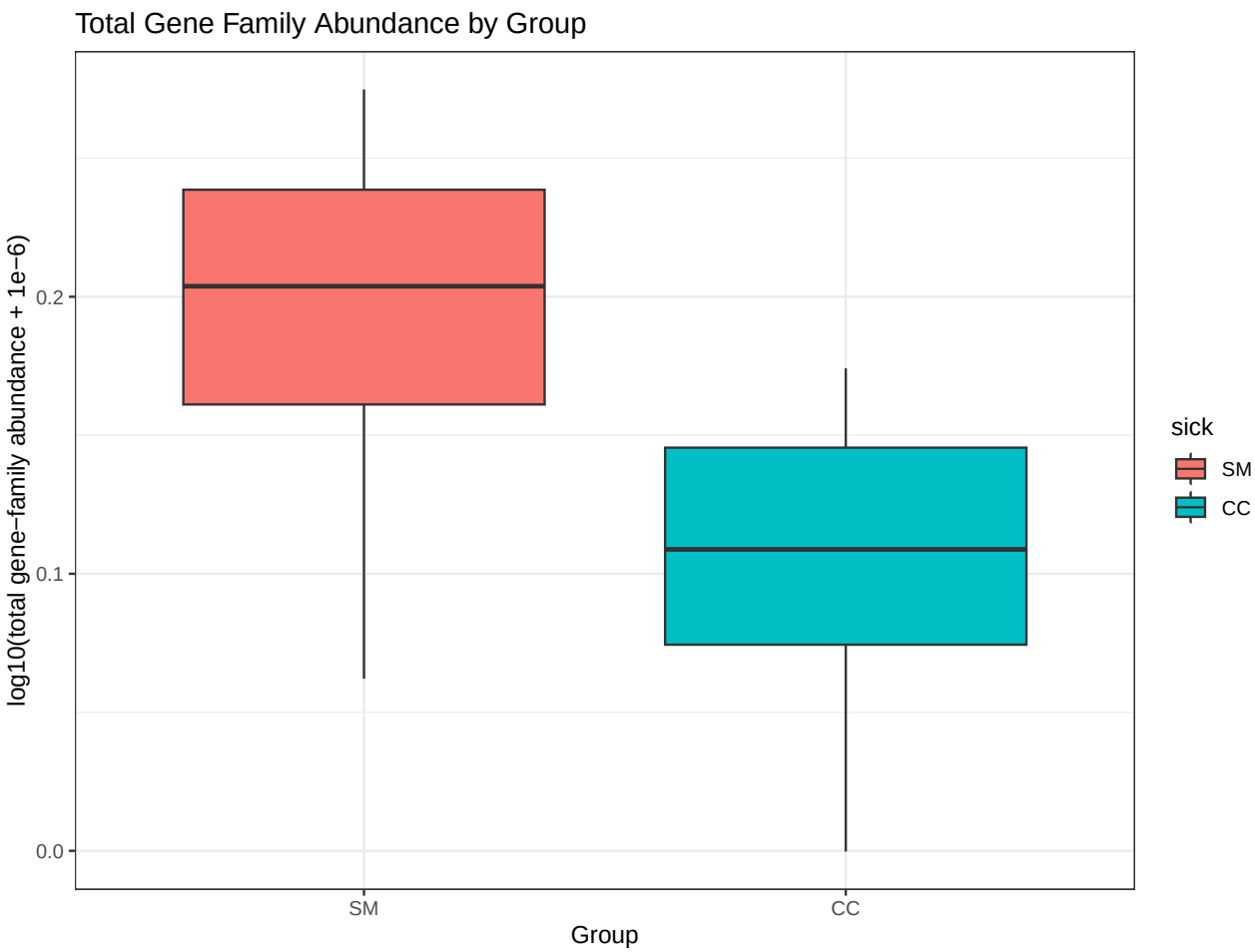
This demonstrates a common feature of microbiome data: increasing resolution improves detail but often reduces interpretability without additional aggregation.

Question 7: Are total pathway or gene-family loads different across groups?

```r
# ==========================================
# GLOBAL FUNCTIONAL LOAD
# ==========================================

meta$total_pathway_abundance <- colSums(path_mat, na.rm = TRUE)
meta$total_gene_family_abundance <- colSums(genefam_mat, na.rm = TRUE)

p1 <- ggplot(meta, aes(x = sick, y = log10(total_pathway_abundance + 1e-6), fill = sick)) +
  geom_boxplot() +
  theme_bw() +
  labs(
    title = "Total Pathway Abundance by Group",
    x = "Group",
    y = "log10(total pathway abundance + 1e-6)"
  )

p2 <- ggplot(meta, aes(x = sick, y = log10(total_gene_family_abundance + 1e-6), fill = sick)) +
  geom_boxplot() +
  theme_bw() +
  labs(
    title = "Total Gene Family Abundance by Group",
    x = "Group",
    y = "log10(total gene-family abundance + 1e-6)"
  )

p1
```


Total Pathway Abundance by Group
log10(total pathway abundance + 1e−6)
−1.10
−1.15
−1.20
−1.25
−1.30
SM
CC
Group
sick
SM
CC

```
p2
```



Interpretation prompt:
These plots ask whether groups differ in the overall amount of annotated functional signal, before focusing on individual features.

Boxplots of total pathway abundance and total gene-family abundance show substantial overlap between SM and CC groups, with no clear or consistent difference in overall functional load.

While there may be minor variation in median values between groups, the distributions largely overlap, indicating that the total amount of functional signal is broadly similar across samples regardless of group.

This suggests that differences observed in earlier analyses are unlikely to be driven by a global increase or decrease in microbiome functional capacity. Instead, any group-associated differences are more likely due to changes in specific pathways or functions rather than overall abundance.

Question 8: Which pathways are significantly associated with group?

```
# ==========================================
# PREPARE INPUTS FOR MaAsLin3
# ==========================================

path_mat_maaslin <- remove_zero_variance_features(path_mat)
genefam_top_maaslin <- remove_zero_variance_features(genefam_top)
```

```r
path_maaslin <- as.data.frame(t(path_mat_maaslin))
gene_maaslin <- as.data.frame(t(genefam_top_maaslin))

stopifnot(all(rownames(path_maaslin) == meta$sample_name))
stopifnot(all(rownames(gene_maaslin) == meta$sample_name))

meta_maaslin <- meta
rownames(meta_maaslin) <- meta_maaslin$sample_name
meta_maaslin$sick <- as.factor(meta_maaslin$sick)

stopifnot(!any(is.na(meta_maaslin$sick)))
stopifnot(!any(duplicated(rownames(meta_maaslin))))

table(meta_maaslin$sick)
```

```
##
## SM CC
## 28  9
```

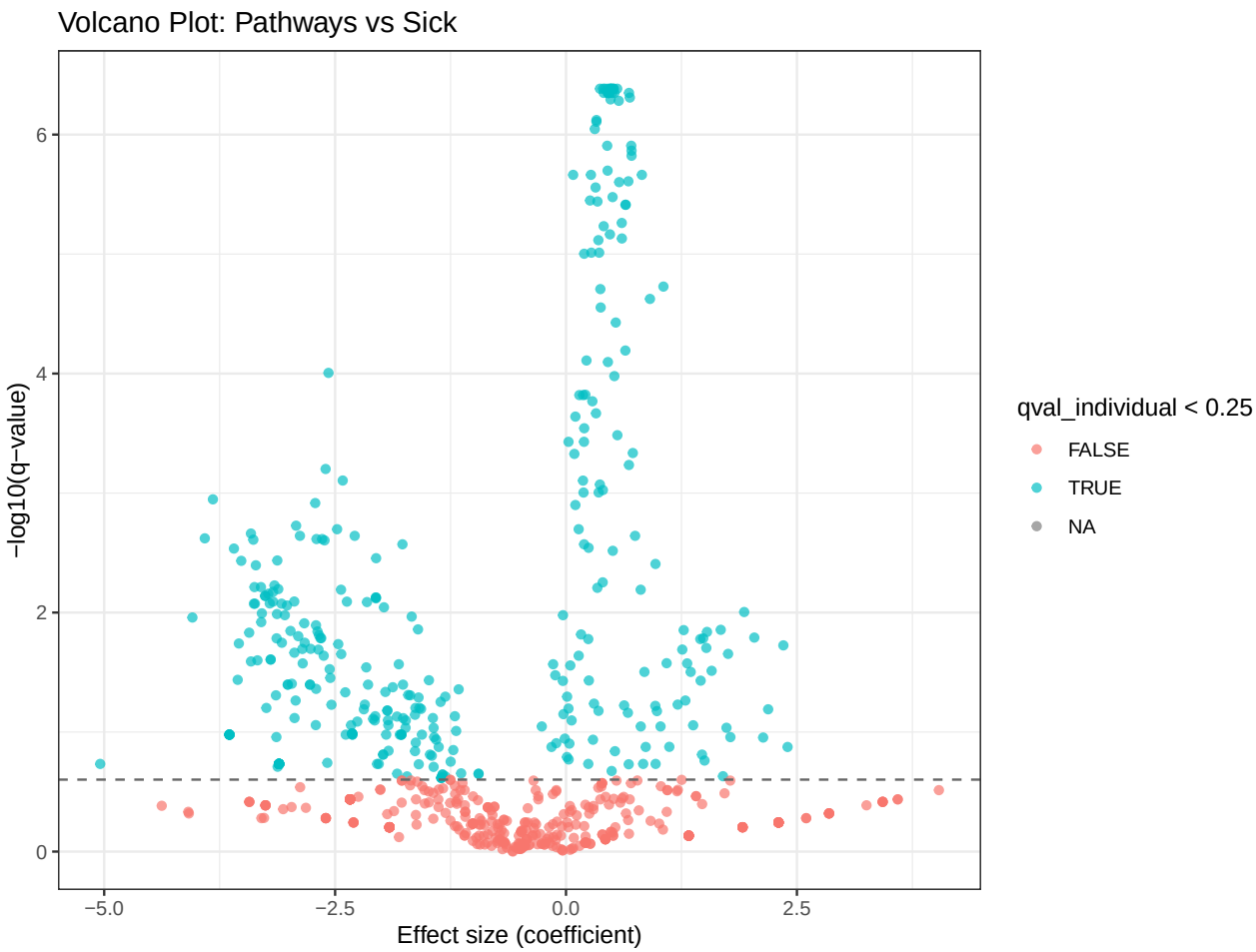
```r
# ============================================
# MaAsLin3: PATHWAYS
# ============================================
# Precomputed externally; shown here for reproducibility only
library(maaslin3)
fit_path <- maaslin3(
  input_data = path_maaslin,
  input_metadata = meta_maaslin,
  output = "maaslin_pathways_group",
  fixed_effects = c("sick"),
  normalization = "NONE",
  transform = "LOG"
)
```

Note: These models are pre-run. Results are read from the output folders.

```r
# ============================================
# MaAsLin3: GENE FAMILIES
# ============================================
# Precomputed externally; shown here for reproducibility only
fit_gene <- maaslin3(
  input_data = gene_maaslin,
  input_metadata = meta_maaslin,
  output = "maaslin_genefamilies_group",
  fixed_effects = c("sick"),
  normalization = "NONE",
  transform = "LOG"
)
```

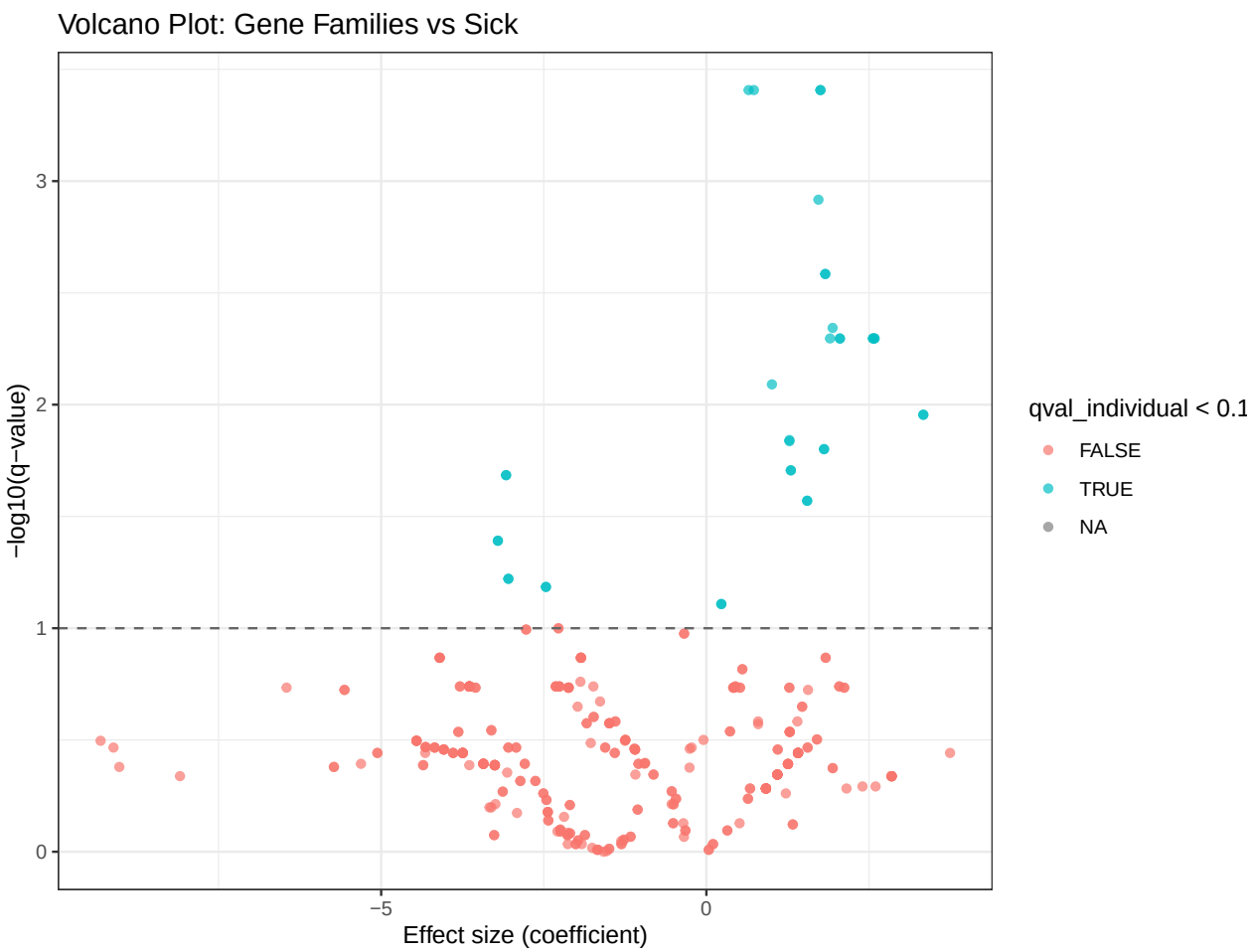
Question 8a: Are there functional differences at the pathway level between the group?

```
# ============================================
# VOLCANO PLOT: PATHWAYS
# ============================================

res_path <- read.delim("maaslin_pathways_group/all_results.tsv")
res_path$clean_feature <- clean_names(res_path$feature)

ggplot(res_path, aes(x = coef, y = -log10(qval_individual))) +
  geom_point(aes(color = qval_individual < 0.25), alpha = 0.7) +
  geom_hline(yintercept = -log10(0.25), linetype = "dashed", color = "grey40") +
  theme_bw() +
  labs(
    title = "Volcano Plot: Pathways vs Sick",
    x = "Effect size (coefficient)",
    y = "-log10(q-value)"
  )
```



Question 8a: Are there functional differences at gene level between the group?

```r
# ==========================================
# VOLCANO PLOT: GENE FAMILIES
# ==========================================

res_gene <- read.delim("maaslin_genefamilies_group/all_results.tsv")
res_gene$clean_feature <- clean_names(res_gene$feature)

ggplot(res_gene, aes(x = coef, y = -log10(qval_individual))) +
  geom_point(aes(color = qval_individual < 0.10), alpha = 0.7) +
  geom_hline(yintercept = -log10(0.10), linetype = "dashed", color = "grey40") +
  theme_bw() +
  labs(
    title = "Volcano Plot: Gene Families vs Sick",
    x = "Effect size (coefficient)",
    y = "-log10(q-value)"
  )
```



Interpretation prompt:

Question 9: Which biologically interpretable pathways appear among the top-ranked (low q-value) results?

```r
# ============================================
# SIGNIFICANT PATHWAY RESULTS
# ============================================

res_path <- read.delim("maaslin_pathways_group/all_results.tsv", check.names = FALSE)

# Clean labels for display only
res_path <- res_path %>%
  mutate(
    clean_feature = gsub("\\.+", " ", feature) %>% trimws()
  )

# Summary first (teaching moment)
cat("Min q-value:", min(res_path$qval, na.rm = TRUE), "\n")
```

```
## Min q-value: Inf
```

```r
cat("q < 0.25:", sum(res_path$qval < 0.25, na.rm = TRUE), "\n")
```

```
## q < 0.25: 0
```

```r
cat("q < 0.10:", sum(res_path$qval < 0.10, na.rm = TRUE), "\n")
```

```
## q < 0.10: 0
```

```r
cat("q < 0.05:", sum(res_path$qval < 0.05, na.rm = TRUE), "\n")
```

```
## q < 0.05: 0
```

```r
# Significant set and annotate biology
sig_path <- res_path %>%
  filter(qval_individual < 0.25) %>%
  arrange(qval_individual) %>%
  mutate(category = annotate_pathways(clean_feature))

# PRINT explicitly
knitr::kable(
  sig_path %>%
    select(clean_feature, category, coef, pval_individual, qval_individual) %>%
    slice_head(n = 20),
  caption = "Top 20 pathways associated with Sick"
)
```

Table 1: Top 20 pathways associated with Sick

clean_feature	category	coef	pval_individual	qval_individual
ANAGLYCOLYSIS-PWY: glycolysis III (from glucose)	Energy metabolism	0.4021688	0	4e-07
BRANCHED-CHAIN-AA-SYN-PWY: superpathway of branched chain amino acid biosynthesis	Amino acid metabolism	0.4948893	0	4e-07
COMPLETE-ARO-PWY: superpathway of aromatic amino acid biosynthesis	Amino acid metabolism	0.5544318	0	4e-07
PEPTIDOGLYCANSYN-PWY: peptidoglycan biosynthesis I (meso-diaminopimelate containing)	Amino acid metabolism	0.4893066	0	4e-07
PWY-5097: L-lysine biosynthesis VI	Amino acid metabolism	0.5162837	0	4e-07
PWY-6385: peptidoglycan biosynthesis III (mycobacteria)	Amino acid metabolism	0.4222933	0	4e-07
PWY-6386: UDP-N-acetylmuramoyl-pentapeptide biosynthesis II (lysine-containing)	Amino acid metabolism	0.4830216	0	4e-07
PWY-6387: UDP-N-acetylmuramoyl-pentapeptide biosynthesis I (meso-diaminopimelate containing)	Amino acid metabolism	0.4755473	0	4e-07
PWY-7953: UDP-N-acetylmuramoyl-pentapeptide biosynthesis III (meso-diaminopimelate containing)	Amino acid metabolism	0.5151027	0	4e-07
PWY-7977: L-methionine biosynthesis IV	Amino acid metabolism	0.3679655	0	4e-07
TRNA-CHARGING-PWY: tRNA charging	Other	0.4531344	0	4e-07
ARO-PWY: chorismate biosynthesis I	Amino acid metabolism	0.5010554	0	4e-07
ILEUSYN-PWY: L-isoleucine biosynthesis I (from threonine)	Amino acid metabolism	0.4516758	0	4e-07
NONMEVIPP-PWY: methylerythritol phosphate pathway I	Other	0.6821428	0	4e-07
PANTOSYN-PWY: superpathway of coenzyme A biosynthesis I (bacteria)	Amino acid metabolism	0.4621764	0	4e-07
PWY-1042: glycolysis IV	Energy metabolism	0.5242867	0	4e-07
PWY-5103: L-isoleucine biosynthesis III	Amino acid metabolism	0.4580145	0	4e-07
PWY-5667: CDP-diacylglycerol biosynthesis I	Amino acid metabolism	0.4603522	0	4e-07
PWY-6609: adenine and adenosine salvage III	Nucleotide metabolism	0.4723834	0	4e-07
PWY-7199: pyrimidine deoxyribonucleosides salvage	Nucleotide metabolism	0.4085923	0	4e-07

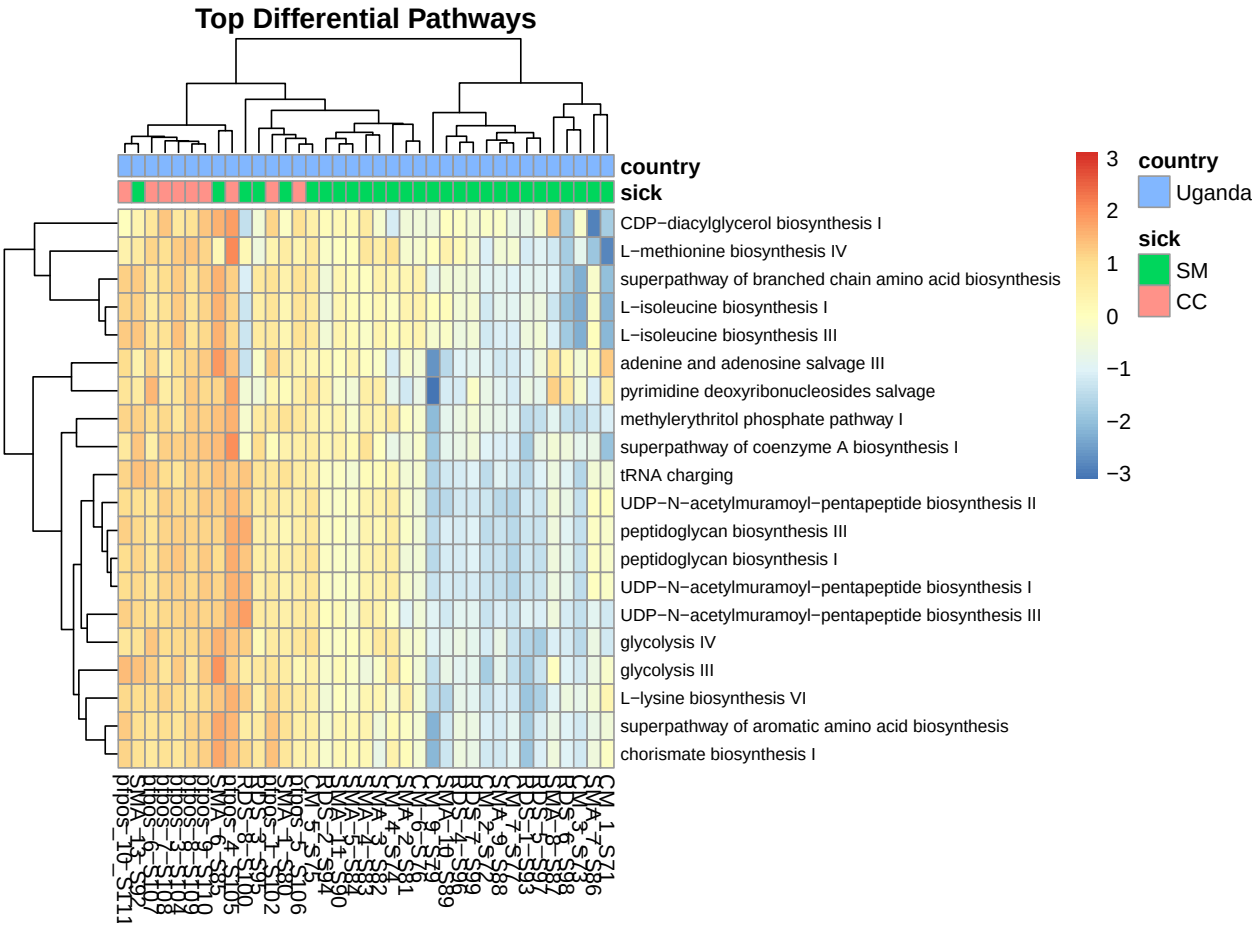
Interpretation prompt:
From summary: q < 0.25: some features and q < 0.10 / 0.05: likely none Therefore:there are no strongly statistically significant pathways, only suggestive associations under a relaxed threshold.

```
# ==========================================
# HEATMAP OF TOP DIFFERENTIAL PATHWAYS
# ==========================================
top_features <- sig_path %>%
  slice_head(n = 20) %>%
  pull(feature)
```

```r
# Try direct matching first (most robust)
top_original <- intersect(top_features, rownames(path_mat))

cat("Mapped features:", length(top_original), "\n")
```

```
## Mapped features: 20
```

```r
if (length(top_original) >= 2) {

  abund_top <- path_mat[top_original, , drop = FALSE]
  rownames(abund_top) <- clean_names(rownames(abund_top))

  anno <- meta %>%
    dplyr::select(sample_name, sick, country) %>%
    as.data.frame()

  rownames(anno) <- anno$sample_name
  anno$sample_name <- NULL

  pheatmap(
    log10(abund_top + 1e-6),
    annotation_col = anno,
    scale = "row",
    fontsize_row = 8,
    main = "Top Differential Pathways"
  )

} else {
  cat("No matching features found - check naming consistency\n")
}
```



Interpretation prompt:
Do the significant features point to broad metabolic shifts, or to specific processes such as respiration, iron handling, or nutrient scavenging?

The top-ranked pathways identified by MaAsLin3, although not statistically significant after multiple testing correction, fall into several biologically interpretable categories. These include amino acid biosynthesis (e.g. branched-chain and aromatic amino acids), central carbon metabolism (glycolysis), nucleotide metabolism (purine and pyrimidine salvage), and cell wall biosynthesis (peptidoglycan pathways).

These functions represent core microbial processes required for growth, replication, and energy production. Their presence among the top-ranked features suggests that variation between groups may primarily involve shifts in fundamental metabolic activity rather than specialised or niche functions.

The heatmap of the top pathways shows modest differences in abundance patterns across samples, with some clustering by group. However, there is substantial overlap, indicating that these pathways are widely shared across the cohort and vary in degree rather than presence or absence.

Importantly, these pathways were selected using a relaxed threshold ($q < 0.25$), and none remain significant under stricter criteria (e.g. $q < 0.05$). Therefore, while these results are biologically interpretable and consistent with known microbial functions, they should be considered exploratory rather than definitive.

Question 10: Can we collapse gene families into targeted biological modules?

This section turns gene families into interpretable functional modules inspired by the paper.

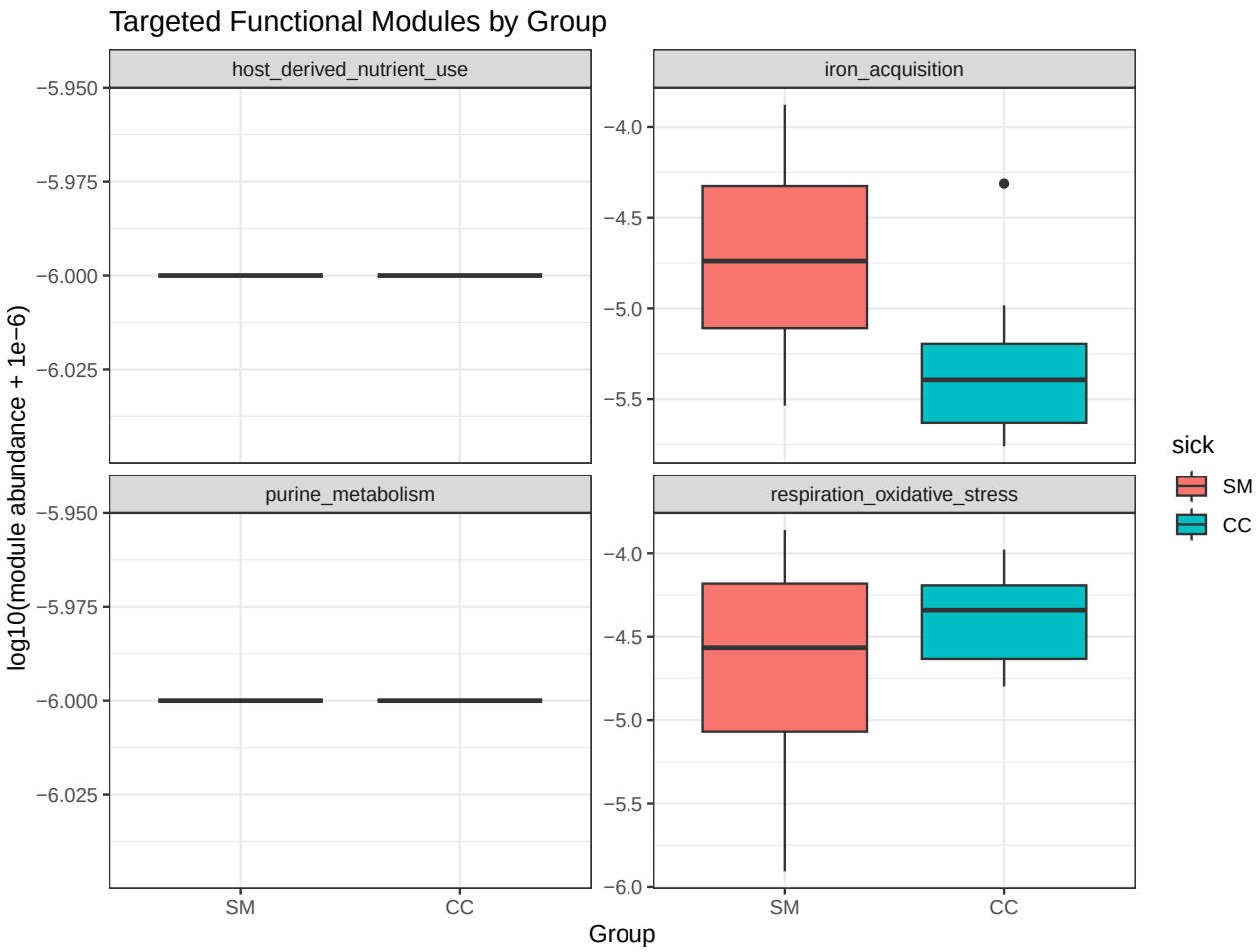
```r
# ============================================
# TARGETED GENE-FAMILY MODULES
# ============================================

gene_ids <- rownames(genefam_mat)

bucket <- rep("other", length(gene_ids))
bucket[grepl("aerobactin|siderophore|fhu|heme", gene_ids, ignore.case = TRUE)] <- "iron_acquisition"
bucket[grepl("formate dehydrogenase|fdn|ubiquinone|menaquinone|superoxide dismutase", gene_ids, ignore.c
bucket[grepl("purine|adenosine deaminase|allantoin", gene_ids, ignore.case = TRUE)] <- "purine_metaboli
bucket[grepl("ethanolamine|fucose|sialic", gene_ids, ignore.case = TRUE)] <- "host_derived_nutrient_use

table(bucket)
```

```
## bucket
##             iron_acquisition                        other
##                           75                      3278448
## respiration_oxidative_stress
##                          116
```

```r
iron_total <- colSums(genefam_mat[bucket == "iron_acquisition", , drop = FALSE], na.rm = TRUE)
resp_total <- colSums(genefam_mat[bucket == "respiration_oxidative_stress", , drop = FALSE], na.rm = TRU
purine_total <- colSums(genefam_mat[bucket == "purine_metabolism", , drop = FALSE], na.rm = TRUE)
hostnut_total <- colSums(genefam_mat[bucket == "host_derived_nutrient_use", , drop = FALSE], na.rm = TRU

target_df <- data.frame(
  sample_name = colnames(genefam_mat),
  iron_acquisition = as.numeric(iron_total),
  respiration_oxidative_stress = as.numeric(resp_total),
  purine_metabolism = as.numeric(purine_total),
  host_derived_nutrient_use = as.numeric(hostnut_total)
) %>%
  left_join(meta, by = "sample_name")
```

```r
# ============================================
# TARGETED MODULES BY GROUP
# ============================================

target_long <- target_df %>%
  pivot_longer(
    cols = c(iron_acquisition, respiration_oxidative_stress, purine_metabolism, host_derived_nutrient_us
    names_to = "module",
    values_to = "abundance"
  )

ggplot(target_long, aes(x = sick, y = log10(abundance + 1e-6), fill = sick)) +
  geom_boxplot() +
  facet_wrap(~module, scales = "free_y") +
  theme_bw() +
  labs(
    title = "Targeted Functional Modules by Group",
    x = "Group",
```

```
  y = "log10(module abundance + 1e-6)"
)
```



Interpretation prompt:
Do these targeted functional modules recapitulate the biology emphasized in the paper?

Among the modules that were detected, iron acquisition appears relatively higher in the SM group than in the CC group. This suggests that microbiome communities in SM may have a greater representation of genes involved in iron scavenging or heme-related acquisition, which is biologically plausible in malaria, where host iron balance, hemolysis, and heme handling are major features of disease.

The respiration/oxidative stress module shows overlapping distributions between groups, with only modest differences. This indicates that genes linked to oxidative stress handling and respiratory processes may vary across samples, but there is no strong or clear separation between SM and CC in this dataset.

Reproducibility

To run this module:

1. Clone the repository
2. Place input files in the `data/` directory
3. Generate the PDF by opening the `.Rmd` file in RStudio
4. Click **Knit → PDF**

Note: MaAsLin3 outputs are precomputed.